

Owen Lee

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EDUCATION

The University of Illinois Urbana-Champaign, Chicago, IL Expected May 2028
Master of Computer Science

The Chinese University of Hong Kong, Shenzhen, Shenzhen, China May 2026
Bachelor of Engineering, Computer Science and Engineering

SKILLS

Programming Languages: Python, Java, C++, Javascript, HTML, SQL
ML Frameworks: PyTorch, HuggingFace, NumPy, scikit-learn
Technologies: Spring Boot, REST APIs, ChromaDB, AWS (EC2, Lambda, Bedrock), Docker, Git, Linux

PROFESSIONAL EXPERIENCE

ATG Nexus Sdn Bhd, Puchong, Malaysia July 2025 - October 2025
Artificial Intelligence Engineering Intern

- Built hybrid keyword+vector retrieval (Weaviate + OpenAI embeddings) for a production legal search app.
- Engineered bilingual triage report generator (Qwen3-Embedding-8B) and Gemini + Plotly Dash pipeline for automated insights, increasing analyst throughput by 50%.

China Unicom, Zhejiang, China May 2024 - August 2024
Software Engineering Intern

- Built a Spring Boot REST API for real-time weather dashboards, sustaining 95% success rate at approximately 2,400 req/hr over 5 weeks in production.
- Implemented server health monitoring with 99% alert detection accuracy, validated against 4 weeks of alert logs.

RESEARCH EXPERIENCE

The Chinese University of Hong Kong, Shenzhen, Shenzhen, China November 2025 - May 2026
Research Assistant (Advisor: Prof. Xiaoying Tang)

- Engineered 1,040 step-grounded visual-text pairs for VLM training, contributing to ICML 2026 acceptance.
- Designed CoT prompting templates aligning rationales with rendered intermediate states, improving training signal for visual reasoning benchmarks.

The Chinese University of Hong Kong, Shenzhen, Shenzhen, China January 2025 - May 2025
Research Assistant (Advisor: Prof. Benyou Wang)

- Built multimodal crawler (1.2M audio-caption pairs) with CLAP-filtered pipeline, cutting hallucination by 15%.
- Trained HTSAT+BERT on EC2-scaled S3 pipeline; achieved 44.3/57.8 R@1 (text→audio / audio→text), best among 5 baselines — contributing to ACL 2026 acceptance.

PROJECTS

Retrieval-Grounded Text-to-SQL Chatbot (ATG Nexus) | Python, MySQL, ChromaDB, SQLGlott October 2025

- Architected schema-aware RAG pipeline (all-MiniLM + ChromaDB) for SQL generation grounded in live DB metadata, achieving 89% execution accuracy on 500+ queries.
- Added SQLGlott guardrails and a 3-try self-repair loop, eliminating unsafe execution.

PUBLICATION

Y. Huo, S. Zhang, K. Zeng, H. Liu, **O. Lee**, J. Chen, Y. Lu, Y. Guo, Y. Liang, X. Tang. *Shape of Thought: Progressive Object Assembly via Visual Chain-of-Thought*. [Paper](#). **International Conference on Machine Learning (ICML), 2026.**

S. Chen, X. Yuan, Z. Li, L. Zhao, **O. Lee**, Z. Su, Q. Sun, B. Wang. *FusionAudio-1.2M: Towards Fine-grained Audio Captioning with Multimodal Contextual Fusion*. [Paper](#). **Association for Computational Linguistics (ACL), 2026.**